mocha+3.0.5

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▼ 1. Overall Concept

The system is built around a **two-layer workflow**:

- **3D view = final result (ground truth centerline)**
- **2D MPR = temporary editing workspace (draft)**

> 👉 Any edits in 2D will NOT affect 3D until you click **“Commit to 3D”**
>
> 👉 Any edits in 3D will immediately update the result, and may sync back to 2D (in Edit mode)

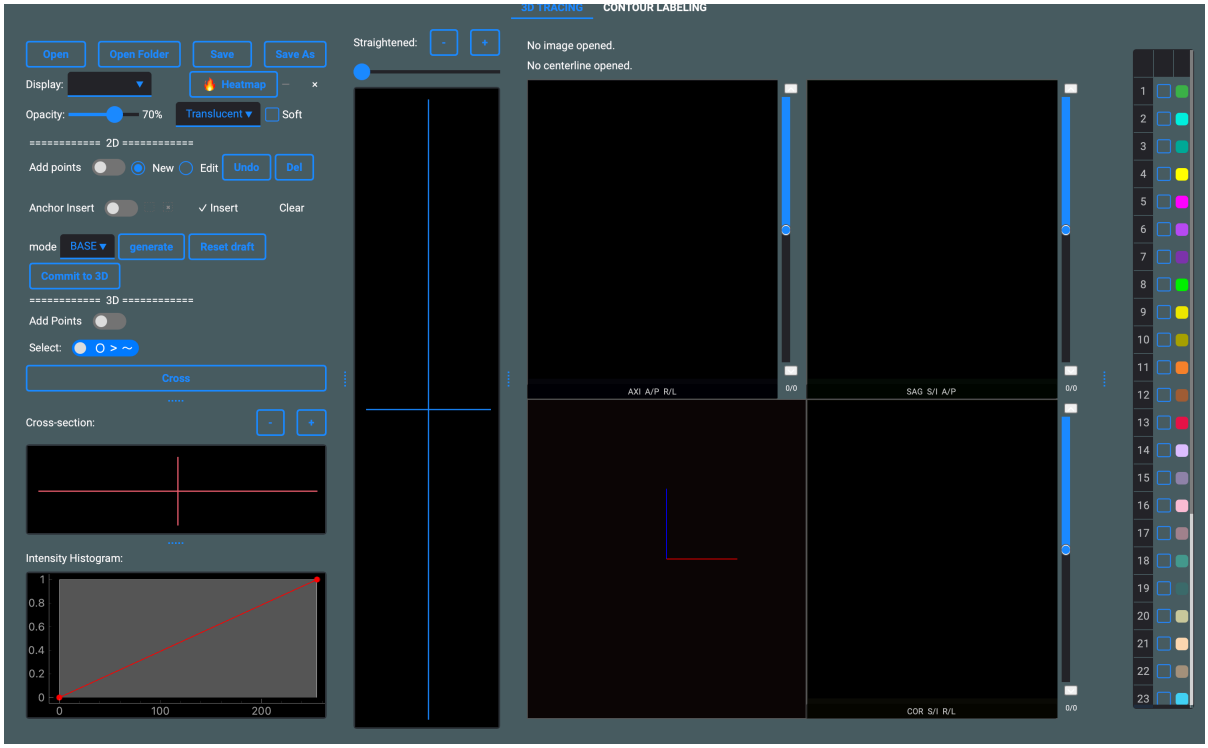
The interface includes:

- 3D view
- Axial / Coronal / Sagittal views

Shortcuts:

- `M`: maximize current view

- `G`: return to grid
- Scroll: change slice
- Shift/Ctrl + scroll: zoom



▼ 2. Data Loading

▼ 2.1 Single Image

1. Click **Open**
2. Select image (NIfTI / DICOM / TIF)
3. Optionally load centerline file

▼ 2.2 Multi-Sequence Folder

1. Click **Open Folder**
2. Select multiple sequences
3. Choose initial display image

4. Switch sequences via dropdown

▼ 3. 3D Editing Workflow

▼ 3.1 Create New Centerline in 3D

1. Enable (3d) **Add Points**
2. Click on vessel in 3D
3. Disable Add Points → segment is created

▼ 3.2 Modify Existing Centerline

1. Click to select
2. Delete key removes points
3. Shift-click adjusts position
4. Cross-section view can refine point position

▼ 4. 2D MPR Editing Workflow

▼ 4.1 Two Modes

Mode	Purpose
New	Create new centerline
Edit	Modify existing

▼ 4.2 Create Centerline in 2D

1. **Switch to New**
2. **Enable Add Points**
3. **Click sparse points in 2D views**
4. **Click Generate**
5. **Click Commit to 3D**

Available tools:

- **Undo**: revert the last action
- **Del**: delete selected point(ctrl/cmd+click)

- **Reset draft**: clear all draft points

Select a fitting mode:

- **BASE**: for bright-blood or standard vessel images
- **VWI**: for dark-blood or vessel wall imaging
- Click **Generate**
- The system automatically fits and sparsifies the path
- Click **Commit to 3D**
- A new segment is written into the 3D centerline

▼ 4.3 Editing an Existing 3D Segment Using 2D (delete point)

1. Select a centerline in 3D (the color→white)
2. Click **Edit** in the 2D panel
3. The selected segment is automatically loaded into the 2D draft
4. Enable **Add points (Edit)**
5. Insert(section 4.5), delete, or adjust points(section 4.4) in 2D views()
6. Click **Generate** for automatic refinement (**optional**)
7. Click **Commit to 3D**
8. The original segment is replaced by the updated one

> 💡 **Note**
>
> - In **Edit mode**:
>   - **Plain click** = insert a new point
>   - **Ctrl/Cmd + click** = select an existing draft point (for deletion)

▼ 4.4 Using Straightened View for Precise Navigation

Workflow

1. Select a centerline segment in the 3D view (it should be highlighted)
2. The **Straightened view** displays the vessel in a flattened representation
3. Click a key point in the straightened view

The system will automatically:

- Move the 2D MPR views to the corresponding slice
- Place the MPR crosshair at that point
- Highlight the same point in 3D
- (In Edit mode) select the corresponding draft point in 2D

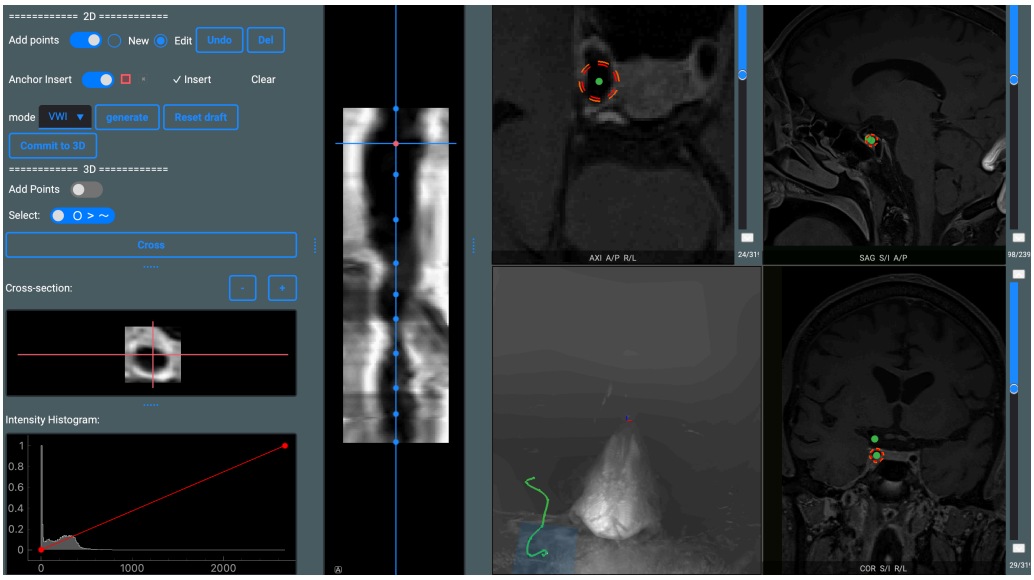
when to use

- Navigating long or tortuous vessels
- Quickly locating a specific region
- Preparing for precise editing in 2D

▼ 4.5 Anchor Insert for Controlled Point Insertion

Workflow

1. Turn **Anchor Insert ON**
2. Configure Anchor Slots
    - **Single-anchor mode**
        - Only enable Slot 1
        - Used for extending from one side
    - **Double-anchor mode**
        - Enable both Slot 1 and Slot 2
        - Used for inserting between two points



3. Select Anchor Points
    - the **Straightened view**, click points to assign anchors
4. Confirm Insertion Constraint
    - Click ✓ **Insert**
5. Go back to **2D MPR**
    - Click once to add a new point
    - 👉 The new point will be **constrained within the anchor-defined region**
6. **Clear** → reset anchors and remove inserted points

Purpose

Use **Anchor Insert** when:

- You want to insert a point **in a specific location**
- You want to avoid “wrong insertion” that creates abnormal curves
- You are performing **fine correction**

💡 Anchor status (visual feedback):

- Gray → disabled
- Green → enabled but not assigned
- Red → anchor point selected